

ATPL EASA STUDY GUIDE | 032 - PERFORMANCE

Chapter 2: Single-Engine Piston Aircraft Performance (Class B)

Syllabus Source: CAE Oxford (pp. 45-78) | Pass Target: $\geq 90\%$ AviationExam

1. CLASS B AIRCRAFT CERTIFICATION (CS-23)

Class B covers propeller-driven aircraft with a Maximum Certified Take-off Mass of 5,700 kg or less, and 9 passenger seats or fewer:

- Take-Off Distance (TOD to 50 ft): The horizontal distance from brake release to the point where the aircraft reaches a screen height of 50 ft above runway surface, with all engines operating.
- Best Angle of Climb Speed (V_x): The speed providing the maximum gain of altitude per unit horizontal distance (steepest climb gradient). Used for clearing close-in obstacles after take-off!
- Best Rate of Climb Speed (V_y): The speed providing the maximum gain of altitude per unit time (highest ft/min). Used for normal en-route climb to cruising altitude.
- Absolute Ceiling vs Service Ceiling: Absolute Ceiling is where maximum rate of climb drops to zero ($V_x = V_y$). Service Ceiling for single-engine aircraft is defined as the pressure altitude where maximum climb rate drops to 100 ft/min.

2. CLIMB GRADIENT VS RATE OF CLIMB FORMULAS

- Climb Gradient (%): $\text{Gradient (\%)} = (\text{Vertical Rise} / \text{Horizontal Distance}) \times 100 = (\text{Thrust} - \text{Drag}) / \text{Weight} \times 100$
- Rate of Climb (ROC) Master Rule: $\text{ROC (ft/min)} = \text{Gradient (\%)} \times \text{TAS (kt)}$ (approximate rule of thumb: $\text{ROC} \sim \text{Gradient \%} \times \text{True Airspeed}$).

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Class B Single-Engine Factored

Problem: POH un-factored takeoff distance to 50 ft screen height is 400 m on dry paved runway. Step 1: Apply EASA commercial safety factor ($\times 1.25$): - Required Distance = $400 \text{ m} \times 1.25 = 500 \text{ meters}$. Step 2: Dry grass correction (+10%): $500 \text{ m} \times 1.10 = 550 \text{ meters}$. Step 3: Uphill runway slope correction (+2% slope adds approx +10%): $550 \text{ m} \times 1.10 = 605 \text{ meters}$. Step 4: Takeoff is legal only if TORA $\geq$ required factored distance!

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Clearing an Obstacle on Take-Off

SCENARIO (Single-Engine Take-off Problem): A single-engine aircraft takes off from a runway: - Take-Off Distance to 50 ft screen height = 600 meters - An obstacle (tree line) of height 110 ft (33.5 m) is located 1,600 meters from brake release - The aircraft climbs at $V_x = 70$ kt TAS with an average climb gradient of 8.0% QUESTION:

Will the aircraft clear the obstacle, and what will be its vertical clearance margin?

SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Determine the horizontal climb distance from 50 ft screen height to the obstacle: - Obstacle distance from brake release = 1,600 m - Distance to screen height (50 ft) = 600 m - Climb distance = 1,600 m - 600 m = 1,000 meters. Step 2: Calculate the altitude gained over this 1,000 m climb distance: - Climb Gradient = 8.0% = 0.08 - Altitude gained = Distance x Gradient = 1,000 m x 0.08 = 80 meters (approx 262 ft). Step 3: Add the initial screen height (50 ft = 15.2 m) to get total aircraft height: - Total height over obstacle = 15.2 m + 80.0 m = 95.2 meters (312 ft). Step 4: Compare with obstacle height: - Obstacle height = 33.5 m (110 ft) - Vertical Clearance = 95.2 m - 33.5 m = 61.7 meters (approx 202 ft)! FINAL ANSWER: Yes! The aircraft clears the tree line with a safe vertical margin of 61.7 meters (202 ft).

[!] EXAM TRAP (AVIATIONEXAM) - V_x vs V_y Wind Effect Trap

- Airspeed for V_x and V_y are INDICATED AIRSPEEDS (IAS) and are NOT affected by wind! - Headwind STEEPENS the flight path gradient over the ground (helps obstacle clearance). - Tailwind FLATTENS the flight path gradient over the ground (danger of hitting obstacles)!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What is the official definition of 'SERVICE CEILING' for a single-engine piston aircraft?**

- [A]: The altitude where climb rate becomes zero
- [B]: The pressure altitude at which the maximum rate of climb drops to 100 ft/min
- [C]: 10,000 ft
- [D]: The altitude where cabin oxygen is required

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under CS-23 certification, the service ceiling for a single-engine aircraft is reached when the maximum full-power rate of climb decreases to 100 ft/min.

Q2: When flying a single-engine aircraft to clear a tall obstacle immediately after take-off, which climb speed M

- [A]: V_y (Best Rate of Climb)
- [B]: V_x (Best Angle of Climb), because it gives the steepest vertical flight path angle per meter of ground distance
- [C]: V_{no}
- [D]: Maximum cruise speed

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. V_x produces the maximum altitude gain per horizontal distance traveled, maximizing vertical clearance over close-in obstacles.

Q3: What happens to the Best Angle of Climb Speed (V_x) and Best Rate of Climb Speed (V_y) as altitude increases?

- [A]: Vx decreases and Vy increases
- [B]: Vx INCREASES slightly and Vy DECREASES until they meet at the absolute ceiling
- [C]: Both speeds stay perfectly constant
- [D]: Both drop to zero

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. With increasing altitude, Vx (IAS) increases slightly while Vy (IAS) decreases. The altitude where Vx equals Vy is the Absolute Ceiling (rate of climb = 0).